

| Variable / proxy            | Level                                      | Construction                                                                                                                                                                                                                                     | Source                                                                | Used for           |
|-----------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------|
| VehicleShare                | vehicle token x day                        | USD volume of indirect routes where token v is an intermediate divided by total indirect-route USD volume that day.                                                                                                                              | data/empirical/bridge_daily.parquet                                   | RQ1, RQ2, RQ3, RQ4 |
| DirectCostAdvantage         | endpoint pair x vehicle x day x trade size | (direct-route exact-quote output USD - indirect-route exact-quote output USD) / direct-route output USD. The candidate-day measure is the median fraction across common-support endpoint pairs at \$10k; positive values favor the direct route. | data/empirical/route_cost_panel_v2.parquet                            | RQ1, RQ3, RQ5      |
| DirectAvailable             | endpoint pair x day x trade size           | Indicator that the exact-quote engine finds a direct executable route for the endpoint pair at the standard notional.                                                                                                                            | data/empirical/route_cost_panel_v2.parquet                            | RQ1, RQ5           |
| IndirectAvailable           | endpoint pair x vehicle x day x trade size | Indicator that both legs of the indirect route through candidate vehicle v are exact-quote executable at the standard notional.                                                                                                                  | data/empirical/route_cost_panel_v2.parquet                            | RQ1, RQ5           |
| DirectDepth                 | endpoint pair x day x trade size           | Executable direct-route quote-quality proxy: direct output USD divided by trade size. Thin-direct cells are direct-available cells with output below 90% of notional; this is not a direct pool-liquidity measure.                               | data/empirical/route_cost_panel_v2.parquet                            | RQ1, RQ5           |
| VehicleLinkedLiquidity      | vehicle token x day                        | Allocated USD TVL in valid Uniswap V3 daily-snapshot pools containing candidate v: full pool TVL when v is the only candidate side and half when both sides are candidates.                                                                      | data/exhibits/lp_concentration.parquet                                | RQ2, RQ3, RQ6      |
| LPConcentration             | vehicle token x day                        | VehicleLinkedLiquidity divided by total vehicle-candidate linked liquidity across the vehicle set that day.                                                                                                                                      | data/exhibits/lp_concentration.parquet                                | RQ2, RQ3           |
| LPRepositioning             | vehicle token x day                        | Daily gross/net V3 mint-burn liquidity movement assigned to candidate vehicle pools; near-price variants use the active/near tick state where available.                                                                                         | data/empirical/_lp_repositioning_day_cache/                           | RQ2, RQ5           |
| Stress                      | day or event day                           | Positive part of negative WETH log return from stablecoin-implied WETH prices; event tables use selected large downside days.                                                                                                                    | bridge_daily WETH price; output/empirical/stress_event_definition.pkl | RQ4                |
| SettlementTransferIncidence | matched route unit                         | Indicator that the transaction receipt contains at least one ERC-20 Transfer log matching the intermediate vehicle token.                                                                                                                        | data/empirical/v4_settlement_transfer_detail.parquet                  | RQ6                |
| NettingExposure             | vehicle token                              | One minus V4 SettlementTransferIncidence for vehicle v in the receipt-audited V4 route-unit sample.                                                                                                                                              | output/empirical/p4b_netting_exposure_by_vehicle.pkl                  | RQ6                |
| VehicleLiquidityFactor      | vehicle token x day                        | Leave-one-out average daily log-liquidity change among other pools linked to the same vehicle; paired with a leave-one-out market liquidity factor.                                                                                              | constructed from raw daily pool snapshots                             | RQ7                |